# A Sharp Entropy–Accuracy Threshold for Local Simulation

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**Abstract**

Consider a binary-output behavior with $m$ inputs per party: equal inputs produce a shared fair bit, whereas distinct inputs produce independent fair bits. A centralized simulator realizes this behavior exactly with two bits of seed entropy. For local deterministic decoders driven by a common discrete seed, we prove that, at every fixed worst-input total variation error $0 < \varepsilon < 1/4$, the minimum seed entropy is

$$(1 - 4\varepsilon) \log_2 \log_2 m + O_\varepsilon(\log_2 \log_2 \log_2 m).$$

At error $1/4$, a seed of entropy less than 1.812 bits suffices uniformly in $m$. Thus the unbounded entropy cost of locality survives every fixed error below an explicit sharp threshold. The proof applies to arbitrary nonuniform countable seeds. Historical priority of the sharp coefficient is not asserted.

## 1 Model and statement

All entropies and unqualified logarithms are in base two. Write $[m] = \{1, \dots, m\}$, with $m \geq 2$. Outputs take values in $\{-1, 1\}$.

Define the target behavior $W_m$ by

$$W_m(a, b \mid i, j) = \begin{cases} \frac{1}{2}\, \mathbf{1}_{\{a=b\}}, & i = j, \\ \frac{1}{4}, & i \neq j. \end{cases}$$

A local simulator consists of a discrete random variable $S$, independent of the inputs, and deterministic functions

$$A_i = f_i(S), \qquad B_j = g_j(S).$$

No communication is allowed. All randomness used by the simulator is included in $S$.

For distributions on a finite alphabet, set

$$d_{\mathrm{TV}}(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)|.$$

Define

$$C_{\mathrm{loc}}(m, \varepsilon) = \inf \left\{ H(S) : \max_{i,j \in [m]} d_{\mathrm{TV}}\big(\mathcal{L}(A_i, B_j), W_m(\cdot, \cdot \mid i, j)\big) \leq \varepsilon \right\}.$$

The infimum permits finite or countably supported seeds.

**Theorem 1** (Sharp entropy–accuracy law)**.** *For every fixed* $0 < \varepsilon < 1/4$*,*

$$\boxed{C_{\mathrm{loc}}(m, \varepsilon) = (1 - 4\varepsilon) \log \log m + O_\varepsilon(\log \log \log m)} \qquad (m \to \infty).$$

*In particular,*

$$\lim_{m\to\infty} \frac{C_{\mathrm{loc}}(m,\varepsilon)}{\log\log m} = 1 - 4\varepsilon.$$

*For every* $\varepsilon \geq 1/4$,

$$\sup_{m\geq 2} C_{\mathrm{loc}}(m,\varepsilon) \leq 1 + h_2(1/4) < 1.812,$$

*where*

$$h_2(p) = -p\log p - (1-p)\log(1-p).$$

## 2 A lower bound for arbitrary seed weights

**Lemma 1** (Heavy-atom obstruction)**.** *Let* $X_1, \ldots, X_m \in \{-1, 1\}$ *be deterministic functions of a discrete seed* $S$*. Suppose*

$$\mathbb{E}[X_i X_j] \leq \rho \qquad (i \neq j),$$

*where* $0 \leq \rho < 1$*. For every integer* $K \geq 1$*, put*

$$r_K = \left\lceil \frac{m}{2^K} \right\rceil.$$

*Then*

$$H(S) \geq (1-\rho)\left(1 - \frac{1}{r_K}\right)\log(K+1).$$

*Proof.* The claim is immediate if $H(S) = +\infty$. Order the seed atoms by nonincreasing mass, $w_1 \geq w_2 \geq \cdots$, appending zero-mass atoms if necessary. Let

$$M_K = \sum_{s=1}^{K} w_s.$$

Each $X_i$ has a sign pattern on the first $K$ atoms. There are at most $2^K$ such patterns. Hence a set $G \subseteq [m]$ of size $r \geq r_K$ has a common pattern. On those first $K$ atoms,

$$\left(\frac{1}{r}\sum_{i\in G} X_i\right)^2 = 1.$$

Nonnegativity elsewhere gives

$$M_K \leq \mathbb{E}\left(\frac{1}{r}\sum_{i\in G} X_i\right)^2 \leq \frac{1}{r} + \rho\frac{r-1}{r}.$$

Consequently,

$$1 - M_K \geq (1-\rho)(1-1/r) \geq (1-\rho)(1-1/r_K).$$

For $s > K$, monotonicity of the weights implies $w_s \leq 1/(K+1)$. Thus, with $0\log(1/0) = 0$,

$$H(S) \geq \sum_{s>K} w_s \log\frac{1}{w_s} \geq (1 - M_K)\log(K+1).$$

Combining the inequalities proves the result. □

**Lemma 2** (Lower bound under simulation error)**.** *For $0 \le \varepsilon < 1/4$ and every integer $K \ge 1$,*

$$C_{\mathrm{loc}}(m, \varepsilon) \ge (1 - 4\varepsilon) \left(1 - \frac{1}{\lceil m/2^K \rceil}\right) \log(K + 1).$$

*In particular, for $m \ge 4$,*

$$C_{\mathrm{loc}}(m, \varepsilon) \ge (1 - 4\varepsilon)(1 - m^{-1/2}) \log\left(\left\lfloor \frac{1}{2} \log m \right\rfloor + 1\right).$$

*Proof.* Consider any admissible simulator. On diagonal inputs, the target assigns zero probability to unequal outputs, so

$$\mathbb{P}(A_j \ne B_j) \le \varepsilon.$$

For distinct inputs, the target correlation is zero. The total variation bound therefore gives

$$|\mathbb{E}[A_i B_j]| \le 2\varepsilon.$$

For $i \ne j$,

$$\begin{aligned} \mathbb{E}[A_i A_j] &= \mathbb{E}[A_i B_j] + \mathbb{E}[A_i (A_j - B_j)] \\ &\le 2\varepsilon + \mathbb{E}|A_j - B_j| \\ &= 2\varepsilon + 2\mathbb{P}(A_j \ne B_j) \\ &\le 4\varepsilon. \end{aligned}$$

Apply the heavy-atom lemma to $X_i = A_i$ and $\rho = 4\varepsilon$.

For the second assertion take $K = \lfloor \frac{1}{2} \log m \rfloor$, so that $\lceil m/2^K \rceil \ge \sqrt{m}$. □

## 3 An asymptotically matching construction

**Lemma 3** (A small approximate synchronous simulator)**.** *For $0 < \eta < 1/2$, define*

$$N = \left\lceil \frac{\ln(2m^2) + 1}{2\eta^2} \right\rceil.$$

*There exists a local simulator $T_{m,\eta}$ with seed entropy $1 + \log N$ such that:*

1. *every output marginal is exactly fair;*
2. *equal inputs give equal outputs almost surely;*
3. *distinct-input correlations have absolute value at most $2\eta$.*

*Its worst-input total variation error relative to $W_m$ is at most $\eta$.*

*Proof.* Sample an $N \times m$ matrix of independent uniform signs. For a fixed pair of distinct columns $i, j$, Hoeffding's inequality gives

$$\mathbb{P}\left(\left|\frac{1}{N} \sum_{s=1}^{N} R_{si} R_{sj}\right| > 2\eta\right) \le 2e^{-2N\eta^2}.$$

A union bound over unordered pairs bounds the probability of any violation by

$$m(m-1)e^{-2N\eta^2} < 1.$$

Fix a matrix with no violation.

Let $J$ be uniform on $[N]$ and let $U$ be an independent uniform sign. Use seed $(J, U)$ and outputs

$$A_i = UR_{Ji}, \qquad B_j = UR_{Jj}.$$

These have the stated properties and seed entropy $1 + \log N$.

A binary pair with fair marginals and correlation $c$ has law

$$\mathbb{P}(A = a, B = b) = \frac{1 + ab\,c}{4}.$$

Its total variation distance from the uniform law is $|c|/2$. This proves the final assertion. □

*Proof of the theorem.* The lower-bound lemma gives, for fixed $\varepsilon < 1/4$,

$$C_{\mathrm{loc}}(m, \varepsilon) \geq (1 - 4\varepsilon) \log\log m - O(1).$$

For the upper bound, fix $0 < \eta < \varepsilon$ and put $v = \varepsilon - \eta$. Let $E_+$ be the local behavior in which both parties always output the same fair sign, and let $E_-$ be the behavior in which they always output opposite fair signs. Each uses one bit of seed entropy.

Consider the mixture

$$L = 3vE_+ + vE_- + (1 - 4v)T_{m,\eta}.$$

Its weights are nonnegative and sum to one. All marginal distributions are fair.

On diagonal inputs its correlation is

$$3v - v + (1 - 4v) = 1 - 2v,$$

so its total variation error is $v$.

For $i \neq j$, write $c_{ij}$ for the correlation of $T_{m,\eta}$. The mixture correlation is

$$2v + (1 - 4v)c_{ij}.$$

Since $|c_{ij}| \leq 2\eta$, its total variation error is at most

$$v + (1 - 4v)\eta \leq v + \eta = \varepsilon.$$

Thus $L$ is admissible.

Choose the mixture mode as part of the seed and then use the corresponding conditional seed. The chain rule yields

$$\begin{aligned} H(S) &\leq H(3v, v, 1 - 4v) + 1 + (1 - 4v) \log N \\ &\leq \log 3 + 1 + (1 - 4v) \log N. \end{aligned}$$

Put $\ell = \log\log m$ and, for sufficiently large $m$, take $\eta = 1/\ell < \varepsilon$. The definition of $N$ gives

$$\log N = \ell + 2\log \ell + O(1), \qquad 1 - 4v = 1 - 4\varepsilon + \frac{4}{\ell}.$$

It follows that

$$H(S) \leq (1 - 4\varepsilon)\ell + O_\varepsilon(\log \ell).$$

Together with the lower bound, this proves the asymptotic.

Finally, the mixture

$$\frac{3}{4}E_+ + \frac{1}{4}E_-$$

has correlation $1/2$ on every input pair. Its error relative to $W_m$ is exactly $1/4$ both on and off the diagonal. A mode bit of entropy $h_2(1/4)$ and an independent fair sign realize it, proving

$$C_{\mathrm{loc}}(m, \varepsilon) \leq 1 + h_2(1/4) \qquad (\varepsilon \geq 1/4).$$

□

## 4 Structural consequence

**Corollary 1** (Robust unbounded locality cost)**.** *A centralized simulator realizes $W_m$ exactly with two bits of seed entropy, for every $m \geq 2$. Nevertheless, for every fixed $\varepsilon < 1/4$,*

$$\sup_m C_{\mathrm{loc}}(m, \varepsilon) = +\infty.$$

*Consequently, no universal additive constant bounds local approximate seed entropy by centralized exact seed entropy, even within this family of classically local behaviors.*

*Proof.* Let $U, V$ be independent fair signs. A centralized decoder, which sees both inputs, returns $(U, U)$ when $i = j$ and $(U, V)$ otherwise. Its seed entropy is two bits. Two bits are also necessary for exact centralized simulation, because any off-diagonal output pair is uniform on four outcomes.

For $0 < \varepsilon < 1/4$, divergence follows from the theorem. For $\varepsilon = 0$, it follows either from the lower-bound lemma or from the exact pairwise-independence bound discussed below. □

## 5 Scope and relation to prior work

At zero error, diagonal agreement identifies $A_i = B_i$ almost surely, while off-diagonal constraints make these $m$ fair bits pairwise independent. The classical entropy bounds imply

$$\log(m + 1) \leq C_{\mathrm{loc}}(m, 0) \leq \lceil \log(m + 1) \rceil;$$

see Gavinsky and Pudlák [1]. The upper bound also follows directly by taking distinct nonzero linear forms of a uniform vector over $\mathbb{F}_2$.

Small probability spaces supporting almost independent bits are classical; see Alon, Goldreich, Håstad, and Peralta [2]. Accordingly, neither the exact logarithmic lower bound nor the existence of double-logarithmic approximate constructions is claimed as new here.

The specific assertion established in this note is the sharp coefficient $1 - 4\varepsilon$ for the stated local simulation problem with arbitrary seed weights, together with its threshold at $\varepsilon = 1/4$. Its historical priority requires a separate literature assessment.

This is a classical locality constraint, not a Bell violation: $W_m$ itself admits exact local realizations. The result does not contradict a centralized causal simulation theorem. It changes the admissible decoder class by restricting each party to its own input.

The construction is existential, using a probabilistic argument. No efficient search algorithm for the sign matrix is claimed. The asymptotic holds for fixed positive $\varepsilon$; its remainder is not asserted to be uniform as $\varepsilon \downarrow 0$. The proof above has not been formally verified in Lean.

## References


[1] D. Gavinsky and P. Pudlák, *On the joint entropy of d-wise-independent variables*, Commentationes Mathematicae Universitatis Carolinae 57(3), 333–343 (2016). doi:10.14712/1213-7243.2015.169.

[2] N. Alon, O. Goldreich, J. Håstad, and R. Peralta, *Simple constructions of almost k-wise independent random variables*, Random Structures & Algorithms 3(3), 289–304 (1992). doi:10.1002/rsa.3240030308.